ANSWER KEY

QP Code 25901119

MAHATMA GANDHI UNIVERSITY, KOTTAYAM

MGU-UGP (HONOURS) REGULAR EXAMINATION March 2026

Fourth SEMESTER

Discipline Specific Elective (DSE) - MG4DSEBOT200, MG4DSEBBT200 - FOOD SCIENCE AND QUALITY CONTROL

(2024 ADMISSION ONWARDS)

Duration: 2 hours **Maximum Marks: 70**

Part A

Short answer

Answer any **10** questions

Each question carries **2** marks

1. What role does fermentation play in traditional diets?

Fermentation plays a foundational role in traditional diets by acting as an ancient method to preserve food, enhance nutrient availability and boost digestibility. By relying on beneficial microorganisms, it converts perishable raw materials into safe, long shelf-life processed food that are heavily integrated into cultures worldwide **(any 2 points)**

2. Identify the mineral that helps in proper muscle contraction.

Calcium or Potassium – full credit (1 mark can be given for Sodium or magnesium)

3. Why are carotenoids more stable than chlorophyll during cooking?

Carotenoids are more heat-stable than chlorophyll during cooking because they lack a central metal ion and possess a rigid, highly stable hydrocarbon structure. But chlorophyll is highly heat-labile, as heating causes its central magnesium atom to be easily displaced, resulting in pigment degradation and colour change.

4. Name an adulterant found in poultry products.

Steroids / Hormones / Formalin (Formaldehyde) (Or any other valid adulterant)

5. Mention any two disadvantages of natural sweeteners used in food.]

- Caloric Content : Many natural sweeteners (honey, jaggery, maple syrup) contain significant calories and can contribute to weight gain if consumed excessively.
- Effect on Blood Sugar: Some natural sweeteners can raise blood glucose levels, making them unsuitable for people with diabetes when consumed in large amounts.
- Sweeting effects of natural sweeteners are difficult to optimize in food products.

- It has less shelf-life compared to artificial sweeteners.
- Are non-economic compared to artificial sweeteners. **(Any Two points)**

6. Name any two common adulterants found in milk.

Water, Starch (Or any valid adulterants in milk)

7. Give two examples of common food adulterants.

Milk- Water, starch

Turmeric- Metanil yellow

Chilli powder- Brick powder

Coffee- Chicory

Salt- Sand (Marks may be given to any two suitable examples)

8. What happens to fruits during moisture loss?

During moisture loss, the water activity in the fruit decreases which inhibits the growth of bacteria, yeast and molds. Moreover, it increases the shelf-life of the fruit.

9. Describe how smoking acts as a preservation method.

Smoking is the traditional preservation method that involves drying and deposition of antimicrobial, antioxidant compounds from wood smoke, such as phenol, formaldehyde and organic acids.

10. Name one control point in quality control.

Inspection of raw materials, Sampling, Testing etc. **(Any one)**

11. What does the FSSAI logo indicate?

The FSSAI logo on a food product shows that it is approved by the Food Safety and Standards Authority of India and meets mandatory safety and quality standards.

12. Name one principle of HACCP.

Identify hazards and assess their risks, Identify the CCPs, Monitor CCPs, establish corrective actions, establish verification process, establish documentation. **(Any one)**

Part B

Short Essay

Answer any **8** questions

Each question carries **5** marks

13. Explain the difference between fat soluble and water soluble vitamins with examples.

Vitamins are classified into fat-soluble vitamins and water-soluble vitamins based on solubility.

Fat Soluble Vitamins:

- Include Vit. A, D, E, K
- Are usually found in fat rich foods and accumulate in body gradually.

- Doesn't have nitrogen in its chemical structure.
- Are mostly heat stable.
- Absorbed indirectly by intestine along with dietary fats and lipids.

Water Soluble Vitamins:

- Include Vit. C and B complex
- They do not accumulate in body as excess is excreted via kidneys.
- B complex has nitrogen in its chemical structure.
- Are less heat stability.
- Absorbed directly into bloodstream.

14. Why are microbial sources preferred for clean label products and large scale industrial applications?

Microbial sources are preferred for clean label products and large-scale industrial applications because they offer sustainable, consistent, and efficient production of bioactive compounds without relying on synthetic chemicals or animal-derived inputs. Microorganisms such as bacteria, yeast, and fungi can be cultivated rapidly in controlled environments, ensuring reproducibility, scalability, and cost-effectiveness. They naturally produce enzymes, vitamins, organic acids, and biofertilizers that align with consumer demand for minimally processed, eco-friendly, and safe products. Additionally, microbial processes reduce environmental impact by lowering resource consumption and waste generation, making them ideal for industries seeking both clean label compliance and industrial efficiency.

15. Explain the importance of carbohydrates in human nutrition.

Carbohydrates are the main energy-giving macronutrients composed of carbon, hydrogen and oxygen. They are the most important and readily available source of energy in the human diet.

Carbohydrates provide about 4 kcal/g of energy and act as the primary fuel for brain and muscle activities. They are the precursors of many fats and amino acids. They store excess energy as glycogen in liver and muscles to meet immediate energy demand of the body. Carbohydrates spare proteins from being used for energy and help in proper fat metabolism and prevent ketosis. They improve digestion through dietary fibre, maintain normal blood glucose level and support overall metabolic activities of the body.

16. Explain food adulteration and its impact on human health.

Food adulteration is the purposeful addition of substances that are foreign, inferior, harmful and unauthorized, to food or removal of valuable components from food to attain unfair profit. This is usually done for various reasons: to gain more profit, increase shelf life, boost volume, improve appearance, enhance taste, and other properties. **(1 mark)**

Impact on Human Health: (4 marks)

Acute Toxicity & Gastrointestinal Issues: Adulterants like contaminated water in milk or chemical additions can cause immediate vomiting, diarrhea, abdominal pain, and severe food poisoning.

Organ Damage & Chronic Toxicity: Long-term consumption of toxic adulterants places a heavy burden on the liver and kidneys (the primary detoxifying organs), leading to liver cirrhosis, kidney stones, or renal failure. For instance, argemone oil in mustard oil causes epidemic dropsy and cardiac failure.

Carcinogenic Risks: Non-permitted synthetic food colors (e.g., metanil yellow in turmeric) and industrial dyes are highly carcinogenic and can lead to various types of

cancers.

Nutritional Deficiencies: Diluting essential foods (like adding water to milk or starch to ghee) reduces the bioavailability of essential macro and micronutrients, leading to malnutrition, especially in children. **(Any four impacts)**

17. Differentiate between stabilizers and thickeners with suitable examples.

Stabilizers	**Thickeners**
Stabilizers help maintain the uniform dispersion of ingredients and prevent separation.	Thickeners increase the viscosity or thickness of food products.
Their primary function is to improve stability and texture.	Their primary function is to improve consistency and body.
They prevent settling of particles and separation of emulsions.	They make foods more viscous and resistant to flow.
Commonly used in ice creams, salad dressings, and dairy products.	Commonly used in soups, sauces, gravies, puddings, and jams.
Examples: Xanthan gum, Carrageenan, Guar gum.	Examples: Starch, Pectin, Gelatin.

Mark Distribution (5 Marks):

- Any **4 valid differences** – 4 Marks
- Suitable examples – 1 Mark

18. Explain food poisoning and its common causes.

Food poisoning is an illness caused by consuming food or beverages contaminated with harmful microorganisms, toxins, chemicals, or other hazardous substances.

Common causes

a) Biological contaminants: Bacteria, virus, parasites etc.
b) Chemical hazards: Pesticide residues, Industrial pollutants
c) Physical contaminants: Shreds of glass, metal etc.

(Food poisoning explanation: 2 marks, Any 3 causes with suitable explanation: 3 marks)

19. Define 'Antioxidant Preservatives' and explain two different ways they protect food quality.

Antioxidant preservatives are natural or synthetic substances added to food to delay or prevent deterioration caused by chemical reactions with oxygen. They primarily act by neutralizing free radicals, thereby protecting the food's nutritional value, flavor, color, and overall shelf life. Egs: Vitamin C, E, citric acid.

1.Preventing Lipid Oxidation (Rancidification)

Unsaturated fats and oils in foods—such as meats, nuts and cooking oils—react easily with oxygen, which causes them to spoil and develop an unpleasant, rancid odor and taste. Antioxidant preservatives (like Tocopherols) readily sacrifice their own molecules to react with oxygen, protecting the fats from breaking down and keeping the food fresh.

2. Inhibiting Enzymatic Browning

When fresh fruits and vegetables are cut and exposed to oxygen, naturally occurring enzymes begin to cause undesirable browning. Antioxidant preservatives (like Ascorbic Acid / Vitamin C) inhibit these enzymes and alter the chemical environment, preventing the formation of brown pigments and maintaining an appealing visual appearance

20. Classify foods based on susceptibility to spoilage and give suitable examples.

Based on susceptibility, foods can be classified into 3 types.

1. Stable or non-perishable foods

   These foods don't spoil easily under normal conditions and stay safe for long periods unless handled carelessly. Example: Sugar, flours

2. Semi perishable food

   These foods can stay fresh for a long period if they are properly handled and stored in appropriate conditions. Example: Potatoes certain varieties of apple

3. Perishable foods

   These are the foods that spoil quickly unless properly preserved. They usually have high moisture and nutrients that support microbial growth. Example: Meat, fish, milk

21. Discuss the importance and procedure of inspection in food industries.

- **Ensures Food Safety**: Inspection identifies contamination, adulteration, and spoilage, preventing unsafe food from reaching consumers.
- **Maintains Quality Standards**: Regular checks help food industries maintain consistent product quality and comply with legal standards.
- **Protects Public Health**: By catching hazards early, inspections help prevent foodborne illnesses.
- **Builds Consumer Trust**: Reliable inspection systems enhance the reputation of food businesses and boost consumer confidence.
- **Legal Compliance**: Inspections ensure adherence to regulations set by authorities like FSSAI, avoiding penalties or shutdowns.

22. Describe the symptoms and control measures of ergotism.

Toxic condition caused by consumption of grains contaminated with ergot alkaloids produced by Claviceps purpurea. **(1 Mark)**
Symptoms – 2 Marks (Any two or more)
Burning sensation in limbs
Muscle cramps and convulsions
Hallucinations
Headache and dizziness
Gangrene due to reduced blood circulation
Control measures – 2 Marks (Any two or more)
Use certified disease-free seeds

Crop rotation
Removal of contaminated grains
Field sanitation
Quality testing before consumption.

Part C

Essay

Answer any 1 questions

Each question carries 10 marks

23. Define lipids. Classify fatty acids into saturated and unsaturated fatty acids with examples and functions.

Lipids are a group of hydrophobic (water-insoluble) organic compounds mainly composed of carbon, hydrogen and oxygen. They include fats, oils, waxes, phospholipids and sterols. Lipids provide a concentrated source of energy, act as structural components of cell membranes, provide insulation and protection to organs and serve as precursors of hormones. **(2 marks)**

Fatty acids are long chain carboxylic acids that form the basic building blocks of lipids.

Classification of fatty acids

Saturated fatty acids: Saturated fatty acids contain only single bonds between carbon atoms so they are fully saturated with hydrogen atoms. They are generally solid at room temperature. Major sources are animal fats like butter, ghee, meat fat and plant oils such as coconut oil and palm oil. They mainly provide energy storage and structural roles, but excess intake may increase cholesterol levels. Examples include palmitic acid and stearic acid. **(4 marks)**

Unsaturated fatty acids: Unsaturated fatty acids contain one or more double bonds in their carbon chain, making them less saturated with hydrogen. They are usually liquid at room temperature. Major sources are vegetable oils (sunflower, olive, soybean oil), nuts, seeds, and fish oils. They help in maintaining membrane fluidity, reducing cholesterol levels, supporting brain and nerve function and acting as essential fatty acids in the diet. Examples include oleic acid, linoleic acid, and linolenic acid. **(4 marks)**

24. Describe high-temperature preservation methods used in food processing.

High temperature preservation is a method of food preservation in which heat is applied to destroy microorganisms and inactivate enzymes responsible for food spoilage. Heat causes proteins and metabolic enzymes of microorganisms to coagulate, thereby preventing their growth and extending the shelf life of foods. The effectiveness of heat treatment depends on the type of microorganism present, the nature of the food, and the preservation method used. **(Definition 2 Marks)**

1. Pasteurization

Pasteurization involves heating food to a temperature below 100°C for a specific period to destroy pathogenic microorganisms while retaining the food's flavour and nutritional quality. It is commonly used for milk, fruit juices, and beverages. A common method is HTST pasteurization, where milk is heated to 72°C for 15 seconds and rapidly cooled.

2. Boiling

Boiling is the heating of food in water at 100°C. It destroys vegetative cells of bacteria, yeasts, and molds but may not kill bacterial spores. It is widely used for cooking foods such as rice, vegetables, meat, and milk.

3. Sterilization

Sterilization involves heating food above 100°C for a sufficient period to destroy all microorganisms, including bacterial spores. It provides long-term preservation and is commonly used in canning and the production of shelf-stable foods.

4. Blanching

It is the preliminary heat treatment performed before freezing or drying. Its primary purpose is to deactivate enzymes that cause browning and quality deterioration, helping to preserve colour, texture and flavour of foods. **(2 Marks each)**